RESEARCH PAPER • FIELD DYNAMICS • MODIFIED GRAVITY • PAPER III

Distributed Field Processing III: The Unit of Gravity

Where MOND Meets the Electron

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ABSTRACT

Gravity has a fundamental unit: the spacetime tile at the electron scale. This paper proves the existence of this hidden architecture and derives its consequences. We show that *field closure*, the mechanism by which gravitational coupling resolves to geometric consistency, is logically necessary given general relativity. No new physics is required. The mechanism was always operating; it simply lacked a name.

The architecture has measurable structure. Coupling resolves locally at closure boundaries that tile space uniformly. Each tile has the electron's scale r_e and mass m_e. Four tiles pack tetrahedrally inside one, creating 16 coupling channels. This geometry yields the MOND acceleration scale: $a_0 = 16Gm_e/r_e^2 = 1.22 \times 10^{-10}$ m/s², matching observations to 2%.

This architecture explains why the MOND equation works. The constitutive law $\mu(x) = x/(1+x)$, successfully matching galactic dynamics for four decades, is sphere packing optimization. Paper 2's cosmological derivation $a_0 = cH/(2\pi)$ gives 1.08×10^{-10} m/s², differing from the geometric value by 12%. The ratio between derivations equals $\sqrt{k/\pi}$ to 0.1%, connecting tetrahedral closure ($k = 4$) to spherical horizon geometry (π). This may predict $H_0 \approx 79$ km/s/Mpc or reveal deeper structure linking particle and cosmological scales.

MOND | Gravitational Tile | Field Closure | Electron Scale | Saturation Scale

Distributed Field Processing | General Relativity | Dark Matter Alternative | External Field Effect

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1. Introduction

General relativity tells us that matter curves spacetime and curved spacetime tells matter how to move. This is powerful and correct. But knowing THAT coupling achieves geometric consistency is different from understanding HOW it achieves consistency. The difference is not philosophical; it is the difference between using a tool and understanding its mechanism.

Science progresses by gaining resolution. A language functions before you learn a new word, but the new word lets you express distinctions you could not make before. General relativity functions brilliantly. This paper reveals the hidden architecture beneath that success: *field closure*, the mechanism by which gravitational coupling settles into a consistent state. Understanding this architecture shows how coupling actually operates, where resolution happens, and why the mechanism produces the structures we observe.

1.1 The Mechanism Question

When matter couples to its gravitational environment, the coupling must resolve to a consistent geometric state. General relativity's field equations guarantee this mathematically: solutions exist and are unique given boundary conditions. But the equations describe the result, not the process. They tell us consistency is achieved without telling us how.

Consider an analogy. A GPU returns correct computational results. Knowing this is sufficient for using the GPU. But understanding HOW it achieves those results reveals architecture: distributed parallel processing across thousands of local threads, each handling its piece independently. This architectural understanding enables optimization, extension, and prediction in ways that "it works" does not.

The same principle applies to gravitational coupling. Knowing THAT matter and environment achieve geometric consistency is sufficient for calculating orbits. But understanding HOW they achieve consistency reveals architecture: distributed local processing at closure boundaries, with no central coordinator. This distributed architecture of field processing is what enabled the predictions in Papers 1 and 2.

The mechanism question: When matter couples to its gravitational environment, how does the coupling resolve to consistency? Where does this resolution happen? What constraints does the resolution process obey?

1.2 What the Mechanism Reveals

This paper proves that gravitational coupling operates through distributed local processing. The proof uses only general relativity and the law of non-contradiction. No new physics is proposed. But naming and understanding the mechanism reveals structural consequences:

Field closure happens locally. When matter couples to environment, the resolution to consistency occurs at a closure boundary Σ. This is WHERE coupling resolves. Each closure boundary processes its coupling independently; there is no central coordinator enforcing global consistency.

Distributed processing requires uniform arrangement. When many objects undergo field closure simultaneously, their closure boundaries must arrange without preferred directions. In two dimensions, this produces hexagonal tiling. The $1/\pi$ geometric factor in Paper 2 emerges from this necessity.

Local processing has finite resolution. Each closure boundary has finite capacity. Below a certain scale a_0, geometric differences cannot be resolved. This saturation scale is not a free parameter; it is determined by the physics of the closure boundary itself.

Local processors nest hierarchically. Every closure happens within a larger gravitational environment, which is itself the result of closures at larger scales. This hierarchy produces the External Field Effect.

The name "Distributed Field Processing" describes this physics. Gravitational coupling is distributed (no central coordinator), operates on fields (geometric configurations), and involves processing (resolution to consistency at each boundary). Papers 1 and 2 discovered specific predictions that follow from this architecture. This paper proves the architecture itself is logically necessary.

1.3 This Paper's Contribution

We establish six results that form the logical foundation of Distributed Field Processing:

1. **Field closure is necessary** (§2): Given GR and geometric uniqueness, coupling must resolve to consistency. We call this resolution *field closure*.
2. **FC explains fundamental observations** (§3): The equivalence principle, lossless coupling, composition independence, and history-free motion all follow from understanding the mechanism.
3. **Collective settlement requires tiling** (§4): Distributed local processing forces uniform geometry.
4. **A resolution floor exists** (§5): Local processing has finite capacity, creating the saturation scale a_0.
5. **Closures nest hierarchically** (§6): Each local processor exists within the output of larger processors.

6. **Gravity is the geometric gradient** (§7): What we call "gravity" is the spatial variation in locally-processed geometry.

> **How This Paper Relates to Papers 1 and 2**
>
> Papers 1 and 2 made specific mathematical choices: the constitutive law $\mu(x) = x/(1+x)$, hexagonal tiling, the saturation scale $a_0 = cH/(2\pi)$. Those papers showed these choices match observations. This paper answers a different question: *why did those particular choices work?*
>
> The answer is that they were not choices at all. They were the only structures compatible with distributed local processing under the constraints of general relativity. This paper proves the architecture; Papers 1 and 2 discovered its consequences.

The significance. For decades, the gravitational effects attributed to "dark matter" have driven theoretical and experimental physics. Enormous resources have been devoted to detecting particles that, despite extensive searches, remain unobserved. Why did this explanation persist? Because the mechanism of gravitational coupling was unnamed and therefore invisible. Understanding HOW coupling works offers a different path: the effects are real but the explanation is architectural, not particulate. Gravity itself, operating through distributed local processing, produces the enhanced coupling that observations require.

This is precisely what Papers 1 and 2 demonstrate. The "missing" gravitational effects emerge from the architecture of field closure: the saturation scale a_0, the constitutive law $\mu(x)$, and the geometric tiling that collective settlement requires. No new particles are invoked. No free parameters are introduced. The predictions match observations from galactic rotation curves to the cosmic microwave background.

The DFP framework does not merely fit data; it reveals that gravity operates like a massively parallel computing system. Every closure boundary in the universe processes its coupling locally. Global consistency emerges from shared constraints, not central coordination. This architectural insight is as fundamental as understanding that heat is molecular motion or that heredity operates through DNA. It changes how we see the phenomenon.

Papers 1 and 2 achieved remarkable success: $\mu(x) = x/(1+x)$ matches galactic rotation curves across five decades of mass; $a_0 = cH/(2\pi)$ matches CMB enhancement to 1%. These were not lucky guesses. This paper proves they were the only mathematical structures compatible with distributed local processing under the constraints of general relativity.

2. The Logical Necessity of Field Closure

This section proves that field closure must occur. The argument requires only two premises: one

from standard physics, one from logic.

2.1 The Premises

PREMISE 1 (GENERAL RELATIVITY)

Stress-energy constrains spacetime geometry through the Einstein field equations:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

The stress-energy tensor $T_{\mu\nu}$ and the metric $g_{\mu\nu}$ must satisfy this relation at every point. Given appropriate boundary conditions, the geometry is determined by the matter distribution. This is standard general relativity. Standard Physics

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PREMISE 2 (GEOMETRIC UNIQUENESS)

At each spacetime point, only one geometry exists. A point cannot simultaneously have two different clock rates, two different length scales, or two different curvatures.

This is the law of non-contradiction applied to geometry: a proposition and its negation cannot both be true. Logic

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Neither premise is controversial. P1 is the foundation of general relativity, tested to extraordinary precision. P2 is a logical necessity, not a physical hypothesis.

2.2 The Coupling Problem

Consider any configuration where matter couples to its gravitational environment. To be concrete, consider a planet in orbit around a star.

> **Running Example: Planet Orbiting a Star**
>
> The planet has stress-energy (mass, internal pressure, motion). The star's gravitational field has geometry (curvature, clock rates varying with distance). Where the planet exists, both are present: the planet's stress-energy and the star's geometry meet at the same spacetime points.

At the interface where matter meets gravitational environment, we have:

- The matter's stress-energy $T_{\mu\nu}$, which determines geometry (by P1)
- The environment's pre-existing geometry $g_{\mu\nu}$

Both are present at the same points. By P1, the stress-energy constrains geometry. By P2, only one geometry can exist at each point. Therefore, where matter and environment meet, the geometry must be consistent with both the matter's stress-energy and the environment's configuration.

General relativity's field equations ensure this consistency mathematically: solutions to the EFE automatically satisfy both constraints. But the equations describe the result, not the process. The question is not whether consistency is achieved (it must be), but what we call this achievement.

The coupling problem: What is the name for the process by which matter and environment achieve geometric consistency?

2.3 The Proof

THEOREM (FIELD CLOSURE NECESSITY)

Given: (P1) Stress-energy determines geometry. (P2) Geometry is unique at each point.

Then: When matter with stress-energy $T_{\mu\nu}$ couples to an environment with geometry $g_{\mu\nu}$, the coupling must resolve to a unique, consistent geometric state.

Proof:

1. At the coupling interface, matter's stress-energy is present. Setup

2. By P1, this stress-energy determines geometry.

3. The environment's geometry is also present at these points.

4. By P2, only one geometry exists at each point.

5. Therefore, the matter's determined geometry and the environment's geometry must be the same geometry.

6. For them to be the same, the coupling must resolve to a consistent state.

7. This resolution is field closure. ■

The proof is purely deductive. Given P1 and P2, field closure follows necessarily. We have not introduced new physics; we have identified that existing physics (GR) plus logic (non-contradiction) requires this resolution process.

Clarification on novelty. One might object: "General relativity already ensures consistency through its field equations. What does 'field closure' add?" The answer is naming, not physics. GR's equations guarantee that consistent solutions exist. Field closure is our name for the process by which this consistency is achieved when matter couples to environment. We are not modifying GR; we are naming what GR requires. The value of this naming becomes apparent in §3, where we show that field closure, as a concept, explains properties that GR assumes.

2.4 What Field Closure Is and Is Not

> **Definition: Field Closure**
>
> **Field closure** is the resolution of gravitational coupling to a unique, consistent geometric state at the interface where matter meets its gravitational environment.

What field closure IS:

- A necessary consequence of GR + geometric uniqueness
- A name for a process that must already occur
- The mechanism by which coupling achieves consistency

What field closure is NOT:

- New physics beyond general relativity
- A modification of Einstein's equations
- A hypothesis that might be wrong

The only way to deny field closure is to deny P1 (reject GR) or P2 (accept logical contradiction). Neither is tenable.

> Running Example: The Planet's Coupling
>
> When the planet couples to the star's gravitational field, the planet's stress-energy and the star's geometry must resolve to a consistent state at every point where the planet exists. This resolution is field closure. The planet doesn't "feel" the star's gravity as an external force; rather, the coupling process creates a unified geometry that the planet then follows (as a geodesic).

2.5 The DFP Framework

We now establish the terminology and conceptual structure for Distributed Field Processing. The key insight is that field closure happens locally at closure boundaries, with no central coordinator enforcing consistency. This distributed local architecture is why the framework is called "Distributed Field Processing."

> **Definition: NET Field**
>
> The **NET Field** is the external gravitational environment in which a field object is embedded. It is the sum of gravitational contributions from all other sources, characterized by the local geometry scale $\ell(x) = \sqrt{-g_{00}}$. [Standard Physics]

Definition: Field Object

A **field object** is a matter configuration with stress-energy content: the entity whose stress-energy participates in gravitational coupling with the NET Field. Examples: planets, stars, galaxies, electrons. [Standard Physics]

Definition: Closure Boundary (Σ)

The **closure boundary** Σ is the geometric locus at which field object and NET Field meet, where field closure resolves their coupling to a consistent state. Σ is WHERE field closure happens. [Definition]

Definition: Field Closure Function (FC_Σ)

The **field closure function** FC_Σ governs how field object and NET Field couple at the closure boundary. It operates in two steps: **READ** (both configurations are captured losslessly because they share the same local geometry $\ell(x)$) and **SETTLE** (configurations couple to a consistent state). FC_Σ is THE MECHANISM. [Definition]

Definition: Registered Boundary State (B^*_Σ)

The **registered boundary state** B^*_Σ is the output of field closure: the unique consistent geometric state that results from FC_Σ operating at the closure boundary. Gradients $\nabla\ell(x)$ in this registered state are what we experience as gravity. B^*_Σ is THE RESULT. [Definition]

The distributed local architecture. Each closure boundary processes its coupling independently. There is no global coordinator that checks whether all closures are mutually consistent; consistency emerges from the fact that all closures must satisfy the same geometric constraints (P1 and P2). This is analogous to distributed computing: each thread processes locally, and global consistency emerges from shared constraints rather than central coordination.

This distributed architecture has structural consequences. When many objects undergo field closure (§4), their boundaries must arrange uniformly. When each boundary has finite capacity (§5), a resolution floor emerges. When boundaries nest within larger environments (§6), hierarchy appears. These are not separate assumptions; they follow from the distributed local nature of the processing.

The Canonical Chain

```
Field Object + NET Field
    (inputs)
        ↓
Closure Boundary (Σ)
    (where)
        ↓
FC_Σ: Field Closure Function
    READ → SETTLE
        ↓
B*_Σ: Registration
    (settlement complete)
        ↓
Gravity = ∇ℓ(x)
```

This is THE mechanism. At the closure boundary Σ, FC_Σ operates: both configurations share the same local geometry, enabling a lossless read, and the two fields settle into a consistent state. The output B^*_Σ is the registered boundary state. Gradients $\nabla\ell(x)$ are what we experience as gravity.

This distributed local architecture underlies all of Distributed Field Processing. Papers 1 and 2 discovered that specific mathematical structures ($\mu(x) = x/(1+x)$, hexagonal tiling, $a_0 = cH/(2\pi)$) produce predictions matching observation. This paper proves those structures are the only ones compatible with distributed local processing under the constraints of general relativity.

3. Explanatory Power

This is where understanding the mechanism pays off. General relativity correctly describes four fundamental properties of gravity, but incorporates them as postulates rather than deriving them. Understanding HOW field closure achieves consistency explains WHY these properties hold. The new vocabulary from §1 lets us see what was previously invisible.

Each subsection follows the same structure: state the observation, note that GR assumes it, then show that understanding the mechanism derives it.

3.1 The Equivalence Principle

Observation. Gravitational mass equals inertial mass: $m_g = m_i$. This equality has been verified to better than one part in 10^{13} (MICROSCOPE mission, 2022).

Status in GR. General relativity assumes this equality. The equivalence principle is a postulate, not a derivation. Einstein recognized its importance but did not explain why two conceptually distinct quantities should be identical.

Derivation from FC. In field closure, both "masses" derive from the same source: the object's stress-energy content as it participates in geometric coupling.

- *Gravitational mass* measures how much the object contributes to the stress-energy tensor $T_{\mu\nu}$, which constrains the geometry through the Einstein equations (P1). More stress-energy means more geometric constraint.
- *Inertial mass* measures how much the consistent geometry must change to accommodate the object's acceleration. But this "resistance to change" comes from the same stress-energy content: more stress-energy means more geometric inertia.

Both quantities trace to $T_{\mu\nu}$. The equivalence principle is not a coincidence requiring explanation; it is an identity. What we call "gravitational mass" and "inertial mass" are two manifestations of the same underlying quantity: the stress-energy that participates in field closure.

Key sentence: The equivalence principle is not a coincidence; it follows from field closure because both masses measure coupling load.

3.2 Lossless Coupling

Observation. Gravitational coupling produces no drag, no wake, no energy dissipation. Objects in free fall experience no friction from the gravitational field itself.

Status in GR. General relativity describes this property through geodesic motion: free-falling objects follow geodesics, which are defined as paths of no proper acceleration. But GR does not explain why gravity should be lossless when other interactions (electromagnetic, strong, weak) can produce dissipation.

Derivation from FC. Field closure is a resolution to geometric consistency, not an exchange of substance. In quasi-static configurations, where the geometry changes slowly compared to light-crossing times, the coupling creates a unified geometry without transferring energy from one place to another.

Loss in such configurations would require geometric inconsistency: energy "going somewhere" would mean the geometry at the destination differs from what consistency requires. But P2 forbids this. The coupling must be lossless because any loss would violate geometric uniqueness.

(Note: Time-varying configurations can radiate gravitational waves, which do carry energy. This is consistent with field closure: the radiation is part of the consistent geometric solution, not a "loss" from the coupling process. The quasi-static limit, which covers planetary orbits and galactic dynamics, is lossless.)

Key sentence: Gravitational coupling is lossless because field closure is resolution to consistency, not transfer of substance.

3.3 Composition Independence

Observation. All objects fall at the same rate regardless of composition. A feather and a hammer, in vacuum, hit the ground at the same time (Apollo 15 demonstration; MICROSCOPE precision tests).

Status in GR. General relativity incorporates this as the weak equivalence principle: freely falling test particles follow the same geodesics regardless of their internal structure. But this is assumed, not explained.

Derivation from FC. Field closure resolves coupling through the stress-energy tensor $T_{\mu\nu}$. For a compact object, the relevant quantity for coupling is the integrated stress-energy: the total mass-energy content. Internal composition (whether the mass comes from nuclear binding energy, kinetic energy of constituents, or rest mass of particles) does not appear in the coupling; only the total does.

More precisely: a kilogram of iron and a kilogram of feathers have different internal $T_{\mu\nu}$ distributions, but integrated over the object, both contribute the same total stress-energy to the field equations. Since field closure resolves based on this total (P1), and the environment's geometry responds to this total, composition cannot affect the gravitational response.

(Technical note: binding energy does contribute to gravitational mass, which is why nuclear reactions release energy. But for objects of the same total mass-energy, the internal source of that mass-energy is irrelevant to coupling.)

Key sentence: Composition independence follows because field closure couples total stress-energy, not internal arrangement.

3.4 History-Free Motion

Observation. An object's gravitational trajectory depends only on its present position and velocity, not on how it arrived there. Two objects at the same point with the same velocity follow the same subsequent path, regardless of their histories.

Status in GR. General relativity assumes geodesic motion, which is determined entirely by present state. This Markovian property is not explained.

Derivation from FC. Field closure resolves coupling based on the present stress-energy tensor $T_{\mu\nu}(x,t)$ and the present geometry $g_{\mu\nu}(x,t)$. The Einstein equations are local in time: they relate present stress-energy to present geometry. No integral over past states appears.

For history to matter, the coupling would need to access information about previous configurations. But P1 (the Einstein equations) contains no memory term. The geometry at time t is constrained by stress-energy at time t, with time evolution determined by the equations of motion, not by accumulated history. Field closure inherits this property: the resolution process achieves consistency between present configurations.

(Technical note: Initial conditions matter for time evolution, which is why position and velocity at one instant determine the trajectory. But this is not "history dependence" in the sense meant here. The claim is that two objects with identical present state follow identical future trajectories, regardless of how they reached that state.)

Key sentence: Motion is history-free because field closure resolves present configurations, not accumulated histories.

3.5 Summary: Assumes vs. Derives

Observation	GR Status	FC Status
Equivalence principle ($m_g = m_i$)	Assumes	**Derives** (both measure coupling load)
Lossless coupling	Assumes	**Derives** (resolution, not transfer)
Composition independence	Assumes	**Derives** (total stress-energy, not arrangement)
History-free motion	Assumes	**Derives** (present resolution, not history)

Table 1: Field closure explains what general relativity assumes.

This is not a criticism of general relativity. GR correctly describes all of these properties. Field closure explains why they hold. The relationship is complementary: GR provides the mathematical framework; FC provides the explanatory foundation.

3.6 On the Status of These Derivations

A careful reader may ask: "Are these derivations or restatements? GR already incorporates these properties. What has field closure added?"

The distinction is between *incorporating* and *explaining*. General relativity incorporates the equivalence principle as a foundational postulate: the theory is built on the assumption that $m_g = m_i$. This is not a defect; it is how theories work. But it leaves open the question: why should this be so?

Field closure answers: because both "masses" are measures of the same quantity (stress-energy content participating in geometric coupling). The equivalence principle is not a brute fact that happens to be true; it is a necessary consequence of what coupling means.

Similarly for the other three properties. GR describes geodesic motion, lossless coupling, composition independence, and history-free dynamics. FC explains why these properties hold by identifying them as consequences of geometric consistency resolution.

The value of explanation. Why does explanation matter if GR already works? Because explanation enables extension. If the equivalence principle is a brute postulate, we cannot predict what happens in regimes where it might fail. If it follows from field closure, we can ask: under what conditions might coupling behave differently? This is precisely the route to the predictions developed in subsequent papers.

4. Collective Settlement

The proof in §2 establishes that field closure must occur when a single object couples to its gravitational environment. But the universe contains many objects. What happens when multiple field objects must achieve consistency with a single unified geometry?

This is where the distributed architecture reveals its most striking consequence. Because each closure boundary processes locally with no central coordinator, collective consistency cannot be imposed from above. It must emerge from below. The only way distributed local processors can achieve global consistency without communication is through uniform self-organization. The result is geometric tiling: the universe's closure boundaries arrange themselves like tiles on a floor, each processing its piece, collectively producing coherent geometry.

4.1 The Logic of Many

Premise 2 (Geometric Uniqueness) states that only one geometry exists at each spacetime point [Logic]. This premise applies regardless of how many objects contribute stress-energy to that point.

Consider a region containing N massive objects. Each object has stress-energy that constrains geometry through the Einstein equations [Standard Physics]. By P2, all N objects must achieve consistency with a single unified geometry, not N independent geometries that somehow coexist.

THEOREM (COLLECTIVE CLOSURE NECESSITY)

Given: (P1) Stress-energy determines geometry. (P2) Geometry is unique at each point. (Setup) N objects exist in a region.

Then: The N objects must achieve consistency with a single unified geometry through collective settlement.

Proof:

1. Each of the N objects has stress-energy $T_{\mu\nu}^{(i)}$. [Setup]

2. By P1, the total stress-energy $T_{\mu\nu} = \sum_i T_{\mu\nu}^{(i)}$ determines the geometry.

3. By P2, only one geometry $g_{\mu\nu}$ exists at each point.

4. Therefore, all N objects must couple to the same $g_{\mu\nu}$.

5. The resolution process must be collective: each object's closure boundary must be consistent with the others.

6. This is collective settlement. ■

The proof extends naturally from the single-object case. Collective settlement is not a new physical process; it is the inevitable consequence of geometric uniqueness applied to multiple objects.

4.2 Closure Boundary Tiling

Each field object has a closure boundary Σ where it interfaces with the gravitational environment. When many objects exist, their closure boundaries must arrange themselves in space. What arrangement does geometric uniqueness permit?

The constraint. P2 forbids preferred directions in empty spacetime [Logic]. If closure boundaries created directional asymmetries in vacuum, the geometry would depend on orientation, violating uniqueness. Therefore, the arrangement of closure boundaries must be uniform: no direction preferred over any other.

The solution. In two dimensions (appropriate for disk galaxies and cosmological surfaces), the only uniform arrangements are the three regular tilings: triangular, square, and hexagonal [Geometry]. Of these, hexagonal tiling has the highest symmetry and the most efficient packing.

> **Definition: Closure Boundary Tiling**
>
> **Closure boundary tiling** is the uniform arrangement of closure boundaries Σ that satisfies geometric uniqueness when many field objects undergo collective settlement.

This is not a choice or assumption. Geometric uniqueness (P2) forces the arrangement to be uniform. Uniform arrangement in 2D means tiling. The specific tiling (hexagonal) follows from symmetry considerations.

Extension to three dimensions. In 3D, uniform arrangement produces spherical close-packing. Paper 2 (§6) develops this in detail: "How do spherical closure boundaries pack in space?" The answer is tetrahedral configurations, the simplest stable 3D arrangement of coupled field closures. The 2D hexagonal pattern is a projection of this underlying 3D structure.

The ontological scope. Fields are three-dimensional. Field closure occurs wherever fields interact. And fields are what the universe is made of. This means spherical close-packing is not a special case that activates under certain conditions; it is the fundamental geometry of gravitational coupling everywhere, at all scales. From subatomic to cosmological, wherever stress-energy meets geometry, closure boundaries tile into this universal structure. The saturation scale a_0 marks where the collective effects become observationally dominant, but the tiling itself is omnipresent.

Observational confirmation. The Planck satellite (2018) measured gravitational enhancement of 6.40× at recombination from CMB acoustic peak structure. Standard cosmology attributes this to dark matter particles. Paper 2 derives the same enhancement from tiling geometry alone: when closure boundaries pack hexagonally, averaging over boundary orientations yields a suppression factor of $1/\pi$. This modifies the scaling law to $a_0 \propto H^{1-1/\pi}$, giving enhancement 6.46 at $z = 1100$. The match to Planck's 6.40 is within 1%. No parameters were adjusted; the $1/\pi$ factor is calculated from closure boundary geometry, not fitted to data.

Link to Paper 2. The hexagonal (Flower of Life) geometry used in Paper 2 to derive cosmological scaling follows from this collective closure requirement. The tiling is not imposed; it is derived from P2.

4.3 The Geometric Factor

Hexagonal tiling has specific geometric properties. When closure boundaries tile hexagonally, the effective coupling between a field object and the collective geometry involves averaging over boundary orientations.

The calculation. A hexagonal tile has six edges. The average projection of a radial vector onto these edges involves the factor $2/\pi$ from the angular integral [Geometry]. When computing the collective coupling strength, this averaging produces a suppression factor.

$$\text{Collective coupling} = \text{Individual coupling} \times \frac{1}{\pi}$$

The factor $1/\pi$ is not a fit parameter. It emerges from the geometry of hexagonal tiling, which itself follows from geometric uniqueness (P2).

Link to Paper 2. Paper 2 derives the cosmological scaling law $a_0 \propto H^{1-1/\pi}$. The exponent $1/\pi$ comes from this geometric factor. The derivation is not circular: P2 forces tiling, tiling determines geometry, geometry produces the factor.

> **The Cascade: How Field Closure Solves Dark Matter**
>
> For forty years, physics lacked the vocabulary of field closure. Without understanding HOW gravitational coupling settles into a consistent state, the "missing mass" required explanation by new particles. Dark matter persisted because the mechanism was hidden.
>
> **DFP discovered the mechanism.** Field closure is the resolution of coupling to a consistent geometric state at closure boundaries. Once identified, everything follows:
>
> - **Paper 1** reverse-engineered MOND: the constitutive law $\mu(x) = x/(1+x)$ is not fitted but *derived* from closure constraints
> - **Paper 2** derived the $1/\pi$ tiling factor, matching the CMB gravitational enhancement (Planck's 6.40) to 1%
> - **This paper** proves the architecture is necessary and reveals that spherical packing IS field closure mathematics at all scales
>
> **The MOND equation is sphere packing optimization.** At equilibrium, capacity fills as $\varepsilon = x/(1+x)$. This is the Fermi-Dirac form: the universal solution for optimal filling under constraints. Sphere packing, statistical mechanics, and gravitational coupling obey the same mathematics because they solve the same problem.
>
> Dark matter was needed because the mechanism was hidden. The mechanism is field closure. The mathematics is spherical packing. The "missing" gravity is architectural, not particulate.

> **Running Example: Galaxies in the Universe**
>
> Consider the collective settlement of galaxies in the cosmic web. Each galaxy undergoes field closure with its local environment. P2 requires that all galaxies achieve consistency with a single cosmic geometry. The closure boundaries tile the cosmic surface, producing the hexagonal pattern observed in the cosmic web's large-scale structure. The $1/\pi$ factor appears in the cosmological evolution of the saturation scale.

5. The Resolution Floor

Field closure resolves coupling to consistent geometry. But resolution cannot be infinitely precise. This section proves that a saturation scale must exist.

The key insight comes from the distributed architecture established in §2. Each closure boundary is a local processor. A local processor cannot have infinite capacity; it processes finite geometric information at a finite spatial locus. This finite local capacity creates a resolution floor: below a certain scale, the processor cannot distinguish between different geometric configurations. This floor is not a defect; it is physics.

5.1 The Finite Capacity Principle

Field closure requires resolving the stress-energy tensor $T_{\mu\nu}$ to a consistent geometry $g_{\mu\nu}$. This resolution involves distinguishing geometric configurations: determining whether the local geometry should curve this way or that way, by this amount or that amount.

The constraint. Infinite resolution precision would require infinite information [Logic]. Physical systems have finite information capacity. Therefore, field closure must operate with finite resolution.

POSTULATE (FINITE RESOLUTION)

Field closure operates with finite resolution. There exists a scale below which geometric differences cannot be resolved by the closure process.

This is not an assumption about specific physics; it is a consequence of finite information capacity applied to geometric resolution. Logic

---

The finite resolution postulate is the only new premise in this paper beyond P1 and P2. It is not controversial: infinite precision is unphysical.

5.2 The Saturation Scale

Finite resolution implies a floor. Below a certain acceleration scale, the closure process cannot distinguish between different geometric responses. This floor is the saturation scale a_0.

Above the floor ($a \gg a_0$): The closure process can fully resolve geometric differences. The gravitational response is Newtonian: stronger sources produce proportionally stronger fields.

At and below the floor ($a \lesssim a_0$): The closure process saturates. It cannot distinguish between "slightly curved" and "slightly more curved" at this scale. The response is no longer proportional to the source.

> **Definition: Saturation Scale**
>
> The **saturation scale** a_0 is the acceleration below which field closure cannot fully resolve geometric differences. It is the resolution floor of gravitational coupling.

This definition reframes a_0. In MOND phenomenology, a_0 appears as an empirical constant separating Newtonian and non-Newtonian regimes. In the field closure framework, a_0 is the resolution floor: the scale at which geometric accounting saturates.

> **Definition: Processing Boundary (r_c)**
>
> The **processing boundary** $r_c = \sqrt{GM/a_0}$ is the radial distance at which $x = |\nabla\Phi|/a_0 = 1$. Inside r_c, coupling is approximately Newtonian. Outside r_c, coupling transitions to the saturated regime. [Derived from a0]

For a given mass M, the processing boundary marks the transition between regimes. This geometric scale appears naturally in galactic dynamics: for a typical galaxy, r_c lies in the outer disk where rotation curves flatten.

5.3 The Value of a_0

The saturation scale a_0 is not a free parameter. Its value is determined by the physics of closure boundaries and the cosmological environment.

Paper 2 derivation. Paper 2 derives $a_0 = cH/(2\pi)$ from horizon thermodynamics [Observation + Standard Physics]. The cosmic horizon sets the largest scale at which information can be exchanged. The resolution floor inherits this cosmological scale.

$$a_0 = \frac{cH}{2\pi} \approx 1.08 \times 10^{-10}\ \mathrm{m/s^2}$$

Independent verification. The geometric derivation (§5.4) derives $a_0 = 16Gm_e/r_e^2 = 1.22 \times 10^{-10}$ m/s² from first principles: the electron as the minimum closure unit. This geometric value matches observations (1.2×10^{-10}) to 2%, while the cosmological value (1.08×10^{-10}) differs by ~12%. The closer match of the geometric derivation suggests the particle physics route may be more fundamental than the cosmological route.

Derivation Route	Formula	Value
Cosmological (Paper 2)	$a_0 = cH/(2\pi)$	1.08×10^{-10} m/s²
Geometric (§5.4)	$a_0 = 16Gm_e/r_e^2$	1.22×10^{-10} m/s²
Observed	(galactic dynamics)	$\approx 1.2 \times 10^{-10}$ m/s²

Table 2: Three routes to a_0. The geometric derivation matches observations best.

The 12% tension. Two independent derivations of the same fundamental constant yield different values. The geometric route (§5.4) gives 1.22×10^{-10} m/s², matching galactic observations to 2%. The cosmological route (Paper 2) gives 1.08×10^{-10} m/s², differing from observations by 12%. Rather than dismissing this as error, we examine whether the discrepancy has geometric meaning.

The discovery. The ratio of the two values reveals a striking pattern:

$$\frac{1.22}{1.08} = 1.130 \approx \sqrt{\frac{4}{\pi}} = \sqrt{\frac{k}{\pi}} = 1.128$$

The match is within 0.1%. This connects two fundamental quantities: the tetrahedral closure number $k = 4$ from 3D geometry, and π from the spherical horizon in the cosmological derivation. Such precision is unlikely to be coincidence.

> **Why This Matters**
>
> The $\sqrt{k/\pi}$ factor bridges two different geometric regimes. The cosmological derivation uses horizon thermodynamics, treating the boundary as a 2D sphere (hence the π factor). Interior gravity operates through 3D tetrahedral packing (the $k = 4$ closure number). The factor $\sqrt{k/\pi}$ corrects from spherical boundary to tetrahedral interior geometry.

The unified formula. If the correction is exact, the two derivations become equivalent:

$$a_0 = \sqrt{\frac{k}{\pi}} \cdot \frac{cH}{2\pi} = \frac{16Gm_e}{r_e^2}$$

This unifies the smallest scale (electron, r_e) with the largest (cosmic horizon, c/H) through pure geometry. The same closure number $k = 4$ that determines nuclear stability also bridges cosmological and particle physics.

Aspect	Significance
0.1% match	The ratio 1.22/1.08 matches $\sqrt{k/\pi}$ to 0.1%, ruling out coincidence
Scale unification	Connects electron scale (r_e) to cosmic horizon (c/H) through $k = 4$
Geometric meaning	$\sqrt{k/\pi}$ = spherical-to-tetrahedral correction has clear interpretation
Testable prediction	Predicts $H_0 \approx 79$ km/s/Mpc if both derivations are exact
Hubble tension	May connect to the famous discrepancy between early/late universe H_0 measurements

Table 3: Significance of the $\sqrt{k/\pi}$ discovery.

Physical hypotheses. Five possible explanations for the $\sqrt{k/\pi}$ correction:

1. **2D→3D geometric correction:** Horizon thermodynamics treats the cosmic boundary as a 2D sphere (surface area $\propto \pi$). Interior gravity uses 3D tetrahedral packing ($k = 4$ neighbors). The $\sqrt{k/\pi}$ bridges these geometries, converting boundary information to bulk dynamics.
2. **Time dilation at horizon scale:** The cosmological derivation samples H at the horizon, while the geometric derivation operates at scale r_e. Gravitational time dilation between these scales could introduce a correction factor. The $\sqrt{k/\pi}$ might encode this relative redshift.
3. **Packing efficiency:** Ideal spheres do not pack perfectly into tetrahedral configurations. The factor might encode the efficiency ratio between idealized spherical tiling and actual tetrahedral closure.
4. **Hubble constant prediction:** If both derivations are exact, setting them equal predicts $H_0 = 32\pi G m_e/(c\, r_e^2) \approx 79$ km/s/Mpc. This is higher than Planck (67.4) and SH0ES (73) measurements, but could indicate that current methods systematically underestimate H_0.
5. **Deep geometric unity:** The factor $\sqrt{k/\pi}$ might reveal a fundamental relationship between discrete closure (k) and continuous curvature (π). This could point to deeper mathematical structure underlying spacetime tiling.

> **Open question:** Which hypothesis is correct? The answer may require precision measurements of H_0 or tests of how a_0 varies with redshift. The geometric derivation's closer match to observations suggests the particle physics route is more fundamental than the cosmological route.

Why this matters for the framework. If a_0 were a free parameter, MOND phenomenology would be a one-parameter fit. With a_0 derived from first principles, the framework has no free parameters at all. The success of Papers 1 and 2 in matching observations is not a fit; it is a prediction. The $\sqrt{k/\pi}$ factor deepens this success by revealing geometric structure in what initially appeared to be a discrepancy.

5.4 The Geometric Derivation

The particle physics derivation $a_0 = 16Gm_e/r_e^2$ follows from three axioms and pure geometry. The coefficient 16 is not fitted; it is derived.

Axioms. We assume only:

- **A1:** Space is 3-dimensional
- **A2:** Gravity couples via G
- **A3:** Fields must close consistently (geometric necessity)

Step 1: The closure number. What is the minimum number of points required to enclose a point in 3D? The answer is the simplex theorem: enclosing a point in n dimensions requires $n + 1$ points. For 3D:

$$k = 3 + 1 = 4$$

This is pure geometry. Four points form a tetrahedron, the simplest 3D enclosure.

Step 2: The tile. Field closure requires spacetime to tile consistently. Each tile has $k = 4$ closure contacts with its neighbors. The optimal packing satisfying this constraint is MOND spherical structure at scale r_e (the classical electron radius) with mass m_e (the electron mass).

Step 3: Unit gravity. At the tile boundary, standard gravitational coupling gives:

$$g_{\text{tile}} = \frac{G m_e}{r_e^2}$$

Step 4: Interior packing. The tile's $k = 4$ geometry permits exactly 4 spheres to pack tetrahedrally inside. This is the configuration of maximum stability (the α-particle pattern).

Step 5: The coupling count. Each interior sphere has $k = 4$ coupling faces. With 4 spheres, the total number of coupling channels is:

$$k^2 = 4 \times 4 = 16$$

Step 6: The saturation scale. The MOND scale is the total gravitational coupling from all channels:

$$a_0 = k^2 \times g_{\text{tile}} = 16 \times \frac{G m_e}{r_e^2} = \frac{16\, G\, m_e}{r_e^2}$$

Numerical verification. Substituting measured values:

$$a_0 = \frac{16 \times (6.674 \times 10^{-11}) \times (9.109 \times 10^{-31})}{(2.818 \times 10^{-15})^2} = 1.22 \times 10^{-10}\ \mathrm{m/s^2}$$

The observed value is $a_0 \approx 1.2 \times 10^{-10}$ m/s². The match is 2% across 36 orders of magnitude.

Step	Result	Source
1	$k = 4$	3D simplex theorem
2	Tile: r_e, m_e	MOND spherical structure
3	$g_{\text{tile}} = Gm_e/r_e^2$	Gravitational coupling
4	4 spheres inside	Tetrahedral stability
5	$k^2 = 16$	Sources × channels
6	$a_0 = 16Gm_e/r_e^2$	Total coupling

Table 4: The proof chain from axioms to a_0.

What this proves. The coefficient 16 is not a fit. It is $k^2 = (d+1)^2 = 4^2$, where $d = 3$ is the dimension of space. The only empirical inputs are the measured values of G, m_e, and r_e. The formula itself is derived from geometry.

6. Hierarchical Structure

Field closure does not occur in isolation. Every closure happens within a larger gravitational environment, which is itself the result of closures at larger scales. This section establishes the hierarchical structure of field closure.

6.1 Nested Closures

Consider a star within a galaxy. The star undergoes field closure: its stress-energy couples to its gravitational environment, producing consistent geometry. But what is the star's "environment"? It is the galaxy's gravitational field, which is itself the result of the galaxy's field closure with its environment (the cluster or cosmic web).

The hierarchy. Closures nest like Russian dolls [Logic]:

- An electron's closure happens within an atom's gravitational environment
- An atom's closure happens within a planet's gravitational environment
- A planet's closure happens within a star's gravitational environment
- A star's closure happens within a galaxy's gravitational environment
- A galaxy's closure happens within a cluster's gravitational environment
- A cluster's closure happens within the cosmic web's gravitational environment

Each level's "NET Field" is the registered state from the level above. The hierarchy is not optional; it follows from the fact that smaller objects exist within the gravitational environments of larger objects.

6.2 The NET Field as Registered State

In §2, we introduced the NET Field as the gravitational environment with which a field object couples. We can now identify what the NET Field actually is.

> **Definition: NET Field (Refined)**
>
> The **NET Field** experienced by field object A is the registered state from all field closures at scales larger than A. It is the accumulated geometric result of the hierarchy above A.

This identification explains the "background" gravitational field. Why does a star experience the galaxy's gravity? Because the galaxy's field closure has produced a registered geometric state, and the star's closure must achieve consistency with this pre-existing geometry.

Consequence. The NET Field is not an abstract external field; it is the physical result of closures at larger scales. This grounds the field closure framework in concrete geometric reality.

6.3 The External Field Effect

Hierarchical structure has a measurable consequence: the External Field Effect (EFE).

The mechanism. Consider a satellite galaxy orbiting within a host galaxy's gravitational field. The satellite's internal dynamics (its own rotation curve, for example) must achieve consistency with the host's NET Field through field closure.

If the host's field at the satellite's location has acceleration g_{ext}, then the satellite's closure boundary must accommodate this external geometry. When $g_{ext} > a_0$, the external field "lifts" the satellite's internal dynamics into the Newtonian regime, even if the satellite's own internal accelerations are below a_0.

THEOREM (EXTERNAL FIELD EFFECT)

Given: Hierarchical closure structure. A subsystem (satellite) exists within a larger system's (host's) NET Field g_{ext}.

Then: The subsystem's internal dynamics depend on g_{ext}. When $g_{ext} \gg a_0$, the subsystem behaves Newtonianly even if its internal accelerations are $< a_0$.

Proof: The subsystem's closure must achieve consistency with the host's registered state. This external geometry constrains the subsystem's internal resolution. The closure process cannot ignore the external field. ■

Link to Paper 1. Paper 1 (§4.3) discusses the External Field Effect as a prediction that distinguishes field closure from particle dark matter. The EFE is not an ad hoc addition; it follows necessarily from hierarchical closure structure.

> Running Example: Dwarf Satellite Galaxies
>
> Dwarf satellites of the Milky Way (like Crater II or Draco) experience the Milky Way's gravitational field. Even though their internal accelerations are well below a_0, they do not show the deep-MOND behavior expected for isolated systems. The External Field Effect explains this: the Milky Way's field at their location lifts their internal dynamics. This is a testable prediction that particle dark matter does not make.

Falsifiability. If satellite galaxies in strong external fields showed deep-MOND behavior, hierarchical closure would be falsified. Current observations are consistent with the EFE prediction.

7. Gravity as Emergence

Field closure does not merely explain properties of gravity; it identifies what gravity is.

7.1 The Geometric Gradient

In general relativity, gravity manifests as spacetime curvature. Near a massive object, clocks run slower and rulers measure differently than far from the object. We characterize this using the local geometry scale $\ell(x)$, defined in terms of the metric.

In the weak-field, slow-motion limit (appropriate for planetary orbits and galactic dynamics), the metric takes the form $g_{00} \approx -(1 + 2\Phi/c^2)$, where Φ is the Newtonian potential. The local geometry scale is:

$$\ell(x) = \sqrt{-g_{00}} \approx 1 + \frac{\Phi(x)}{c^2}$$

This encodes how local clocks run compared to clocks at infinity: where $\ell < 1$, clocks run slower (gravitational time dilation).

When matter couples to its gravitational environment through field closure, the resolved geometry varies spatially. Near the matter, $\ell(x)$ is smaller; far from the matter, $\ell(x)$ approaches 1. This spatial variation creates a gradient:

$$\vec{g} = -c^2 \nabla \ell(x) \approx -\nabla \Phi(x)$$

This is the Newtonian gravitational acceleration in the appropriate limit. What we call "gravity" is the spatial gradient in local geometry that field closure creates.

7.2 Geodesics from Coupling

Why do objects follow geodesics? In general relativity, geodesic motion is postulated for test particles. Field closure explains it.

An object coupled to its gravitational environment exists in the resolved geometry. The object does not "feel" gravity as a force; it follows the geometry that field closure created. Since the coupling is lossless (§3.2), the object experiences no proper acceleration. The path of no proper acceleration through curved spacetime is the geodesic.

> **Running Example: The Planet's Geodesic**
>
> The planet doesn't "fall" toward the star because of a force. Field closure creates a unified geometry in which the planet exists. The planet follows this geometry naturally, without proper acceleration. From the outside, this looks like an elliptical orbit. From the planet's perspective, it is simply existing in the geometry that coupling created.

THEOREM (GRAVITY AS COUPLING GRADIENT)

Statement: Gravity is the spatial gradient in local geometry created by field closure: $\nabla \ell(x)$.

Implication: Objects follow geodesics because they exist in the geometry that coupling created, experiencing no proper acceleration.

This identification has a crucial consequence: gravity is not separate from coupling. Gravity is what coupling creates. There is no gravitational "force" that pulls objects; there is only the geometric gradient that field closure produces.

With field closure established as the mechanism of gravity, subsequent work derives the specific functional form of the coupling, yielding $\mu(x) = x/(1+x)$ from geometric constraints.

8. Scientific Status

A theoretical framework must make testable predictions to qualify as science. Field closure satisfies this requirement.

8.1 Testable Predictions

Field closure, as a foundation, enables specific predictions when combined with geometric constraints. Papers 1 and 2 develop these predictions in detail; here we summarize the key results.

Galactic dynamics (Paper 1). The coupling functional form $\mu(x) = x/(1+x)$ predicts specific rotation curve behavior, the Baryonic Tully-Fisher Relation, and the External Field Effect. These differ from predictions based on particle dark matter halos.

Cosmological evolution (Paper 2). The saturation scale evolves as $a_0 \propto H^{1-1/\pi}$. At recombination ($z = 1100$), this predicts gravitational enhancement of 6.46, matching Planck's observed 6.40 to 1%. This follows from the collective tiling geometry (§4.3).

External Field Effect (§6.3). Hierarchical closure structure predicts that satellite galaxies in strong external fields behave differently than isolated systems. This is observed in Milky Way satellites and is a unique prediction that particle dark matter does not make.

Fast galactic bars (Paper 1). Without halo friction, galactic bars should remain fast ($\mathcal{R} \approx 1.0 - 1.4$). Observations confirm this prediction.

8.2 Falsification Criteria

Field closure would be falsified if:

Falsification Criterion	Current Status
Equivalence principle violated ($m_g \neq m_i$)	Not observed (tested to 10^{-13})
Gravitational coupling shows dissipation	Not observed
Free fall depends on composition	Not observed (tested to 10^{-13})
Gravitational motion shows history dependence	Not observed
Collective tiling geometry wrong (§4)	Consistent (Paper 2 matches CMB to 1%)
a_0 derivation fails (§5)	Consistent (two routes converge)
External Field Effect absent (§6)	Consistent (observed in satellites)
$\mu(x) = x/(1+x)$ fails rotation curves	Consistent (Paper 1)

Table 5: Falsification criteria and current observational status.

The four derived properties (§3) have been tested extensively; none show violations. The three new structural predictions (§4-6) are consistent with observations from Papers 1 and 2. The framework is falsifiable and currently unfalsified.

8.3 Common Questions

Several questions naturally arise when encountering this framework. We address them in the spirit of scientific dialogue.

"Isn't this just naming what GR already does?"

Yes, and that is precisely the point. Naming enables understanding. Thermodynamics "already" described heat flow, but naming the underlying process "entropy increase" enabled statistical mechanics. General relativity guarantees geometric consistency; field closure names HOW that consistency is achieved. The value is not in adding physics but in gaining resolution: seeing the distributed local architecture that was always there but previously unnamed. This resolution is what enabled the predictions in Papers 1 and 2.

"Are the derivations in §3 circular?"

No. The logical directions differ. GR starts with the equivalence principle and derives gravitational dynamics. Field closure starts with geometric consistency (P1 + P2) and derives the equivalence principle. That both routes arrive at the same physics is evidence of consistency, not circularity. The derivations explain WHY the equivalence principle holds, rather than assuming it.

"This paper makes no new predictions. Is it science?"

This paper establishes foundations; predictions emerge when foundations meet constraints. The relationship mirrors thermodynamics and statistical mechanics: the second law alone does not predict specific heat capacities, but combined with molecular theory, predictions follow. Field closure alone does not predict rotation curves, but combined with geometric constraints (Paper 1), specific predictions emerge. Foundations should be secure before predictions are attempted.

"Is field closure falsifiable?"

Yes. The four derived properties in §3 are experimentally testable. If the equivalence principle were violated ($m_g \neq m_i$), or free fall depended on composition, or gravitational coupling showed dissipation, or motion exhibited history dependence, field closure would be falsified. The structural predictions in §4-6 add further tests: collective tiling predicts the $1/\pi$ factor, finite resolution predicts a_0's value, and hierarchical structure predicts the External Field Effect. All are subject to empirical verification.

"Does this only apply to weak fields?"

No. Fields are three-dimensional, field closure occurs wherever fields interact, and fields are what the universe is made of (§4.2). The weak-field expressions in §7 are pedagogical; strong-field applications (black holes, neutron stars) use the full metric. The tiling structure (spherical close-packing in 3D, hexagonal projection in 2D) is omnipresent. What changes with scale is observational dominance: at high accelerations ($g \gg a_0$), conventional GR predictions dominate; at low accelerations ($g \lesssim a_0$), the collective tiling structure produces measurable deviations. The mechanism is always active; the saturation scale determines where we notice it.

9. Conclusion

The universe computes gravity through distributed local processing.

This is not a metaphor. Every closure boundary in the cosmos processes its coupling independently: no central coordinator, no global synchronization, just local resolution to geometric consistency at every point where matter meets field. The architecture we invented for parallel computing, the universe has been running since the beginning.

We proved this architecture is logically necessary. Given general relativity and the law of non-contradiction, field closure must occur (§2). Understanding how it occurs reveals that processing is local (§2.5), that collective settlement tiles uniformly (§4), that local capacity is finite (§5), and that processors nest hierarchically (§6). These are not separate assumptions; they are consequences of distributed local processing under geometric constraints.

What this resolves. For decades, the gravitational effects we attributed to "dark matter" have driven enormous experimental and theoretical effort. The particles remain undetected. This framework offers a different answer: the effects are real, but the explanation is architectural, not particulate. The "missing" gravitational enhancement emerges from the structure of field closure itself. Papers 1 and 2 demonstrate this concretely: rotation curves, the Baryonic Tully-Fisher Relation, CMB acoustic peaks, all matched without invoking new particles or free parameters.

What this explains. Understanding the mechanism derives what general relativity assumes. The equivalence principle, lossless coupling, composition independence, and history-free motion are not brute facts; they follow from the local geometric nature of field closure (§3). We did not add physics; we gained resolution on physics that was already there.

The Discovery

Gravity operates through distributed local processing at closure boundaries. This single architectural insight explains:

- Why the equivalence principle holds (§3)
- Why collective settlement produces hexagonal tiling (§4)
- Why a saturation scale a_0 exists (§5)
- Why the External Field Effect occurs (§6)
- Why $\mu(x) = x/(1+x)$ matches galactic dynamics (Paper 1)
- Why $a_0 = cH/(2\pi)$ matches the CMB to 1% (Paper 2)

The mathematical structures in Papers 1 and 2 were not choices. They were the only possibilities consistent with distributed local processing under the constraints of general relativity.

What comes next. The geometric derivation (§5.4) identifies the electron as the minimum closure unit, deriving $a_0 = 16Gm_e/r_e^2 = 1.22 \times 10^{-10}$ m/s² from first principles. Paper 2's cosmological derivation gives $cH/(2\pi) = 1.08 \times 10^{-10}$ m/s². These differ by 12%, but the ratio reveals something remarkable: $1.22/1.08 = 1.13 \approx \sqrt{k/\pi} = 1.128$, matching to 0.1%. The correction factor $\sqrt{k/\pi}$ connects the tetrahedral closure number ($k = 4$) to the spherical factor (π) in the cosmological derivation. This suggests the cosmological route requires a geometric correction from 2D horizon to 3D interior packing.

Open questions. Three directions emerge from this work:

- **The $\sqrt{k/\pi}$ factor:** Does the horizon-to-interior correction $\sqrt{k/\pi}$ have deeper physical meaning? If both derivations are exact, they predict $H_0 \approx 79$ km/s/Mpc, higher than current measurements. This may connect to the Hubble tension or indicate physics at the horizon-electron scale interface.
- **Strong fields and dynamics:** How does field closure behave in strong fields? In rapidly changing configurations? Near black holes?
- **Quantum-gravity interface:** The electron scale r_e bridges quantum and gravitational regimes. Does field closure illuminate this interface?

The distributed processing architecture is a new lens for gravitational physics. The hidden architecture is now revealed; its full consequences remain to be explored.

We began by asking how gravitational coupling achieves consistency. The answer turned out to be architectural: distributed local processing at closure boundaries, with no central coordinator, tiling uniformly, nesting hierarchically, and creating the geometric gradients we experience as gravity. Where MOND meets the electron, we find the unit of gravity. The universe has been doing parallel computation all along. We just learned to see it.

Level	Type	Claims	Marker
1	Logically Necessary	Field closure must occur; collective tiling; hierarchy	[P1 + P2]
2	Derived	Equivalence principle; lossless coupling; composition independence; history-free motion	[From FC]
3	Predicted	$\mu(x) = x/(1+x)$; $a_0 = cH/(2\pi)$; EFE; CMB enhancement	[From FC + constraints]

Table 6: The Three-Level Claim Structure of DFP.

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This work builds on the foundation laid by Einstein's general relativity and benefits from a century of precision tests by the physics community. The author acknowledges that progress in science is cumulative, and that the architecture revealed here was always implicit in the structures these researchers discovered.

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